1. Solution Techniques – Introduction

- After the elements are formed, we have to assemble them and solve a large set of simultaneous equations: $[K]\{U\} = \{F\}$.
- Steps involved: (i) numerical integration of element matrices, (ii) condensation of unwanted dof, (iii) assembly of elements, (iv) imposing boundary conditions and (v) solving the equations.
- For static loads the load does not change with time and inertia forces are ignored.

2. Numerical Integration

2.1 Need

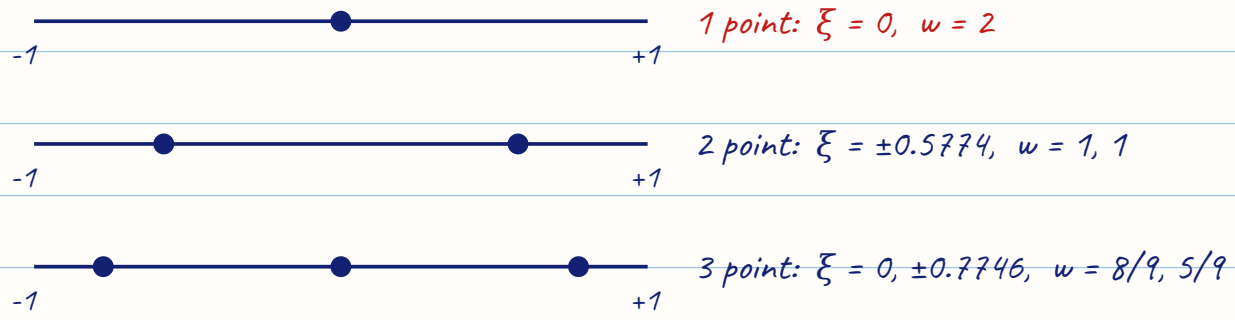
- In isoparametric elements the integrand $[B]^T[D][B]\det[J]$ is a complicated rational function of ξ and η – exact integration is not possible.
- So the integral is replaced by a weighted sum of the integrand values at selected sampling points.

$$\int_{-1}^{1} f(\xi)\, d\xi \approx \sum_{i=1}^{n} w_i f(\xi_i)$$

2.2 Newton–Cotes and Gauss quadrature

- Newton–Cotes (trapezoidal, Simpson): equally spaced points; n points integrate exactly a polynomial of degree (n-1).
- Gauss quadrature: sampling points and weights chosen optimally; n points integrate exactly a polynomial of degree (2n-1). Fewer points needed – hence preferred in FEM.

2.3 Gauss points and weights (limits -1 to +1)



- n = 1: $\xi = 0$, $w = 2$ (exact for linear polynomial).
- n = 2: $\xi = \pm 1/\sqrt{3} = \pm 0.57735$, $w = 1$ each (exact for cubic).
- n = 3: $\xi = 0, \pm\sqrt{(3/5)} = \pm 0.77460$, $w = 8/9, 5/9, 5/9$ (exact for degree 5).
- n = 4: $\xi = \pm 0.33998, \pm 0.86114$, $w = 0.65215, 0.34785$.

2.4 Change of limits

- If the integral is over [a, b], convert to the natural interval [-1, 1] by

$$x = (b - a)/2 \cdot \xi + (b + a)/2, \quad dx = (b - a)/2 \, d\xi$$

$$\int_a^b f(x)\, dx = (b - a)/2 \int_{-1}^{1} f(\xi)\, d\xi$$

2.5 Two-dimensional integration

$$\int_{-1}^{1} \int_{-1}^{1} f(\xi,\eta)\, d\xi\, d\eta \approx \sum_{i=1}^{n} \sum_{j=1}^{n} w_i\, w_j\, f(\xi_i, \eta_j)$$

- 2×2 Gauss rule (4 points): $\xi, \eta = \pm 0.57735$, weights $1 \times 1 = 1$.
- 3×3 Gauss rule (9 points): weights 25/81, 40/81, 64/81.

- Choice of order: Q4 element → 2×2; Q8 / Q9 element → 3×3. Too few points causes zero-energy (hourglass) modes; too many gives no extra accuracy.

2.6 Solved example

- Problem: Evaluate $I = \int_{-1}^{1} (3e^x + x^2 + 1/(x+2))\, dx$ by 2-point Gauss quadrature.

  Points: $x = \pm 0.57735$, $w = 1$.

  $f(0.57735) = 3(1.7813) + 0.3333 + 1/2.57735 = 5.3439 + 0.3333 + 0.3880 = 6.0652$

  $f(-0.57735) = 3(0.5614) + 0.3333 + 1/1.42265 = 1.6842 + 0.3333 + 0.7029 = 2.7204$

  $I \approx 1 \times 6.0652 + 1 \times 2.7204 = 8.7856$
- Exact value: $3(e - 1/e) + 2/3 + \ln 3 = 7.0512 + 0.6667 + 1.0986 = 8.8165$. The error is small (about 0.4%).

3. Static Condensation

3.1 Concept

- Also called substructuring or Guyan reduction. Some dof (usually internal / interior nodes) are eliminated at the element or substructure level, so the global system becomes smaller.
- Dof are divided into master (retained) $\{u_m\}$ and slave (condensed) $\{u_s\}$ dof.

K_{mm}	K_{ms}
K_{sm}	K_{ss}

$$\begin{Bmatrix} u_m \\ u_s \end{Bmatrix} = \begin{Bmatrix} F_m \\ F_s \end{Bmatrix}$$

m = master (retained)

s = slave (condensed)

$$K^* = K_{mm} - K_{ms}K_{ss}^{-1}K_{sm}$$

3.2 Derivation

$$[K_{mm}\ K_{ms}\ ;\ K_{sm}\ K_{ss}]\{u_m\ ;\ u_s\} = \{F_m\ ;\ F_s\}$$

- From the second row: $K_{sm}u_m + K_{ss}u_s = F_s \rightarrow u_s = K_{ss}^{-1}(F_s - K_{sm}u_m)$
- Substitute in the first row: $K_{mm}u_m + K_{ms}K_{ss}^{-1}(F_s - K_{sm}u_m) = F_m$

$$[K^*]\{u_m\} = \{F^*\}$$

$$[K^*] = [K_{mm}] - [K_{ms}][K_{ss}]^{-1}[K_{sm}]$$

$$\{F^*\} = \{F_m\} - [K_{ms}][K_{ss}]^{-1}\{F_s\}$$

- After solving for u_m, the slave displacements are recovered (back-substitution):

$$\{u_s\} = [K_{ss}]^{-1}\ (\{F_s\} - [K_{sm}]\{u_m\})$$

3.3 Advantages

- Reduces the size of the global matrix and the solution time; less storage.
- Useful in dynamic analysis (reducing dof) and for large repeated

substructures (e.g. aircraft wing, ship hull).

- No approximation is made in static analysis – the results for retained dof are exact (Guyan reduction is approximate only for dynamics).

3.4 Solved example

- Problem: Condense out dof 2 from $[K] = k\,[2\ -1\ 0;\ -1\ 2\ -1;\ 0\ -1\ 1]$ (dof 1, 2, 3), keeping dof 1 and 3.

  Master: 1, 3. Slave: 2. $K_{ss} = 2k$, $K_{ms} = k\,\{-1\ -1\}^T$, $K_{mm} = k\,[2\ 0;\ 0\ 1]$

  $[K^*] = k\,[2\ 0;\ 0\ 1] - (k^2/2k)\,[1\ 1;\ 1\ 1] = k\,[1.5\ -0.5;\ -0.5\ 0.5]$

- The reduced 2×2 matrix is symmetric and of reduced size.

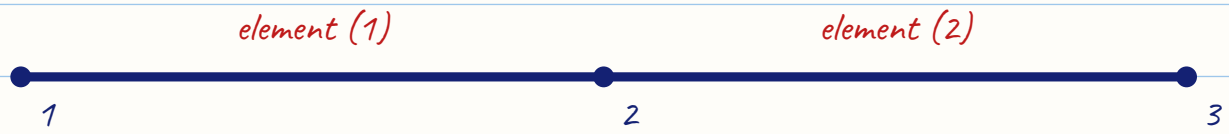
4. Assembly of Elements

4.1 Global stiffness matrix

- Individual element matrices are combined to form the global matrix by the direct stiffness method:

  $[K] = \Sigma_{e=1}^{n}\,[k^{(e)}]$, $\{F\} = \Sigma\,\{f^{(e)}\}$

- The element local dof are mapped onto the global dof using the connectivity (element–node table); each term k_{ij} of the element matrix is added to the global location K_{IJ}.



$[K^{(1)}]$ → rows/cols 1,2 $[K^{(2)}]$ → rows/cols 2,3

overlap at node 2: $K_{22} = k_{22}^{(1)} + k_{11}^{(2)}$

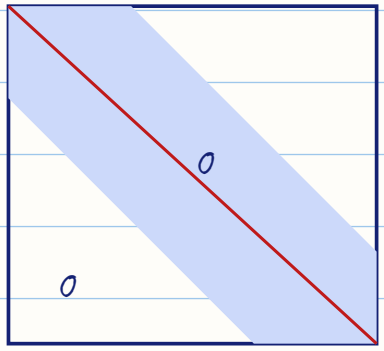
$[K] = \Sigma\ [k^{(e)}]$ (direct stiffness method)

4.2 Steps

- Step 1: Prepare the connectivity table (element, node i, node j) and the global dof numbering.
- Step 2: Initialise $[K] = 0$ of size (total dof)^2.
- Step 3: For each element, compute $[k^{(e)}]$ in the global coordinate system (transform if inclined).
- Step 4: Add $k_{ij}^{(e)}$ to K_{IJ} where I, J are the global dof numbers.
- Step 5: Similarly assemble the load vector.
- Terms at shared nodes are added, other terms are copied.

4.3 Properties of [K]

- Symmetric: $K_{ij} = K_{ji}$. Banded: non-zero terms near the diagonal, due to local connectivity of elements.
- Sparse: most terms are zero. Singular before boundary conditions are applied.
- Positive diagonal terms; positive definite after supports are imposed.



Banded, symmetric, sparse

Only non-zero band is stored

Half-bandwidth: nb = (D+1)·f

D = max node-number difference in an element

4.4 Bandwidth

Half bandwidth $nb = (D + 1)\, f$

- D = maximum difference between node numbers of any element, f = dof per node.
- Efficient numbering of nodes (to reduce D) reduces storage and computing time; number nodes along the shorter direction of the structure.
- Methods: banded storage, skyline (profile) storage, and frontal solution.

4.5 Example – assembly of 3 bars in series

- Nodes 1-2-3-4, elements (1-2), (2-3), (3-4) with stiffnesses k_1, k_2, k_3:

$$[K] = [\, k_1 \; -k_1 \; 0 \; 0 \; ; \; -k_1 \; k_1+k_2 \; -k_2 \; 0 \; ; \; 0 \; -k_2 \; k_2+k_3 \; -k_3 \; ; \; 0 \; 0 \; -k_3 \; k_3 \,]$$

- Half bandwidth for a bar: $nb = (1 + 1) \times 1 = 2$.

5. Boundary Conditions

5.1 Types

- Essential (geometric) boundary conditions: prescribed displacements ($u = 0$ at a fixed support). Must be applied.
- Natural (force) boundary conditions: prescribed loads / tractions – included in the force vector.

5.2 Elimination approach

- Remove the rows and columns of the fixed dof from $[K]$ and $\{F\}$; solve the remaining reduced system. Reactions are calculated afterwards:

  $$\{R\} = [K]\{U\} - \{F\}$$

- For a non-zero prescribed displacement a at dof j: subtract $K_{ij} \cdot a$ from the right side of every other equation i and then delete row j.

5.3 Penalty approach

- Add a very large number C to the diagonal K_{jj} and $C \cdot a$ to F_j.

  $$(K_{jj} + C)\, U_j + \Sigma\, K_{ji} U_i = F_j + C\, a \rightarrow U_j \approx a$$

- Size of the matrix is unchanged, so the programming is easy. Typical $C = 10^4$ to 10^6 times the largest diagonal term.

  Reaction at the support: $R_j = -C\,(U_j - a)$

5.4 Multi-point constraints (short note)

- Relations of the form $\beta_1 U_i + \beta_2 U_j = \beta_0$ (inclined rollers, rigid links) are handled by penalty or Lagrange multiplier methods.

6. Solution Techniques for Static Loads

6.1 Direct methods

- Gauss elimination: reduce $[K]$ to upper triangular form by forward elimination, then back-substitution.
- Cholesky (LDL^T) decomposition: for symmetric positive-definite $[K]$, $[K] = [L][L]^T$; solve $[L]\{y\} = \{F\}$, then $[L]^T\{U\} = \{y\}$. Takes advantage of symmetry and banded form; about half the effort of Gauss elimination.
- Frontal solver: assembles and eliminates equations at the same time, so only the active front is kept in memory.
- Used in most commercial software because they are reliable and give the exact answer (up to round-off).

6.2 Gauss elimination steps

- Step 1: Eliminate U_1 from equations 2, 3, ... n using equation 1:

  $K_{ij}^{(1)} = K_{ij} - (K_{i1}/K_{11})\, K_{1j}$, $F_i^{(1)} = F_i - (K_{i1}/K_{11})\, F_1$
- Step 2: Continue for U_2, U_3, ... to get an upper triangular system.
- Step 3: Back substitution:

  $U_n = F_n^*/K_{nn}^*$, $U_i = (F_i^* - \Sigma_{j>i} K_{ij}^* U_j) / K_{ii}^*$
- Pivot (diagonal) terms must be non-zero and positive for a stable

structure; a zero pivot signals a mechanism (insufficient supports).

6.3 Iterative methods

- Gauss–Seidel, Jacobi, conjugate gradient. Start with an assumed {U} and improve it step by step until the change is small.

$$U_i^{(k+1)} = (1/K_{ii}) \left(F_i - \Sigma_{j<i} K_{ij} U_j^{(k+1)} - \Sigma_{j>i} K_{ij} U_j^{(k)} \right)$$

- Good for very large 3-D problems because only non-zero terms are stored; convergence depends on conditioning of [K].

6.4 Direct vs iterative

- Direct: fixed number of operations, robust, good for many load cases, needs more memory.
- Iterative: less memory, suited to very large sparse systems, convergence not guaranteed.

7. Complete Solution Flow (Static Analysis)

- (1) Pre-processing: geometry, mesh, material properties, loads, supports.
- (2) Element matrices: $[k^{(e)}]$ using shape functions; numerical integration if needed; condense internal dof.
- (3) Assembly: $[K]\{U\} = \{F\}$ using connectivity.
- (4) Apply boundary conditions (elimination or penalty).
- (5) Solve using Gauss elimination / Cholesky / iterative solver.
- (6) Post-processing: displacements, strains $\{\varepsilon\} = [B]\{q\}$, stresses

$\{\sigma\} = [D]\{\varepsilon\}$, reactions; check equilibrium.

- Mesh convergence: refine the mesh until the results (stress, displacement) no longer change significantly (h-refinement), or increase the polynomial order (p-refinement).

8. Solved Example – Assembly, Boundary Conditions and Soluti

- Problem: Two bars in series (nodes 1-2-3), each with AE/l = 1000 N/mm. Node 1 is fixed; loads: 500 N at node 2 and 300 N at node 3. Find the displacements.

  $[K] = 1000\ [1\ -1\ 0\ ;\ -1\ 2\ -1\ ;\ 0\ -1\ 1]$

  After $U_1 = 0$: $1000\ [2\ -1\ ;\ -1\ 1]\ \{U_2\ U_3\}^T = \{500\ 300\}^T$

  Gauss elimination: $R_2 \leftarrow R_2 + 0.5\ R_1 \rightarrow 1000\ (0.5)\ U_3 = 300 + 0.5 \times 500 = 550$

  $U_3 = 550/500 = 1.1$ mm

  $2000\ U_2 - 1000 \times 1.1 = 500 \rightarrow U_2 = 0.8$ mm

- Reaction at node 1: $R_1 = -1000\ (U_2) = -800$ N (balances the total load 500 + 300 = 800 N□).
- Bar forces: $P_1 = 1000(U_2 - U_1) = 800$ N, $P_2 = 1000(U_3 - U_2) = 300$ N.

9. Quick Revision Points

- Gauss quadrature with n points is exact up to polynomial degree 2n-1; weights: 1 point (2), 2 points (1, 1).
- Static condensation: $K^* = K_{mm} - K_{ms}K_{ss}^{-1}K_{sm}$.

- Assembly: add the element terms at common dof; [K] is symmetric, banded and sparse.
- Boundary conditions: elimination or penalty method.
- Solution: Gauss elimination / Cholesky for direct, Gauss–Seidel for iterative.

Important questions: (1) explain Gauss quadrature with example, (2) derive static condensation, (3) assembly of global stiffness matrix with banded form, (4) elimination and penalty approaches, (5) solve a small system by Gauss elimination.